Cyber Security

Unit III: Mobile & Wireless Security

Mobile Device Security & Vulnerabilities

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YEAR : I Year II Semester

ACADEMIC YEAR : 2025 - 2026

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Mobile Device Proliferation & Risks

Mobile Proliferation has transformed corporate operations. Bring Your Own Device (BYOD) policies allow employees to access sensitive corporate databases on personal smartphones, creating severe security challenges.

The BYOD Security Threat

- **Mixed Contexts:** Personal apps (games, social media) run alongside corporate emails and databases on the same device.
- **Unsecured Networks:** Mobile devices connect to public, unencrypted Wi-Fi networks, exposing traffic to sniffing.
- **Physical Loss / Theft:** Smartphones are easily lost. If unencrypted, finder has access to stored company data.
- **Rooted / Jailbroken Devices:** Users bypass OS security rules, enabling malware to access sandbox resources.

Common Mobile Attack Vectors

- **Malicious Apps:** App store clones hiding spyware, tracking locations, and logging keystrokes.
- **Smishing (SMS Phishing):** SMS messages containing links to fake bank portals or malware downloads.
- **Credit Card / SMS Frauds:** Malicious apps registering users to premium SMS services or copying credit card details from tap-to-pay NFC interfaces.
- **Vulnerable OS versions:** Slow vendor update cycles leave Android/iOS devices unpatched against known kernel exploits.

Mobile OS Security & MDM Controls

To secure mobile endpoints, enterprises use a combination of OS features and MDM software:

Mobile OS Security Features

- **App Sandboxing:** Each app runs in an isolated container. A game cannot read data from a banking app's memory.
- **Permission Systems:** Users must explicitly authorize app access to camera, contacts, location, and files.
- **Biometric Authentication:** Uses face/fingerprint scans to unlock cryptographic key chains dynamically.
- **Hardware Encryption:** Storage drives are encrypted by default, utilizing dedicated secure enclave processors.

Mobile Device Management (MDM)

- **Central Administration:** Software (e.g. Microsoft Intune) that manages corporate configurations on devices.
- **Remote Wipe:** Instantly erases all corporate data if a device is reported lost or stolen.
- **Secure Containerization:** Separates personal apps from corporate apps, blocking copy-paste actions between them.
- **Compliance Locks:** Blocks devices from corporate networks if they are rooted, jailbroken, or missing password locks.

Applications of Mobile Security

Where mobile security protocols are implemented to secure transactions and assets:

🏦 Mobile Banking Applications

Uses app sandboxing, mandatory MFA, biometric verification, and SSL pinning. Detects rooted devices to block execution, preventing balance tampering.

🏥 Securing Electronic Health Records (EHR)

Doctors use tablet systems to view patient records. MDM containers ensure HIPAA compliance by encrypting data and blocking screen captures.

🏢 Corporate BYOD Email Access

Companies use MDM policies to lock email access behind secure PIN containers, protecting corporate data on employee phones.

🛒 Mobile Point of Sale (POS) Tablets

Retail tablets running card transactions encrypt swipe data at the card reader, preventing credit card theft over wireless networks.

THANK YOU

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